with

$$Z_{\beta,\mu}(x) = \int e^{\beta\langle x,y\rangle}\,\mathrm{d}\mu(y). \tag{3.3}$$

In other words, (SA) is a *mean-field interacting particle system*. The evolution of $\mu(t)$ is governed by the continuity equation[7]

$$\begin{cases} \partial_t\mu + \operatorname{div}(\mathcal{X}[\mu]\mu) = 0 & \text{on } \mathbb{R}_{\geqslant 0}\times\mathbb{S}^{d-1} \\ \mu_{|t=0} = \mu(0) & \text{on } \mathbb{S}^{d-1} \end{cases} \tag{3.4}$$

satisfied in the sense of distributions.

Remark 3.1 (Well-posedness)**.** *Global existence of weak, measure-valued solutions to* (3.4) *for arbitrary initial conditions* $\mu(0)\in\mathcal{P}(\mathbb{S}^{d-1})$ *follows by arguing exactly as in* [GLPR24, Lemma A.3]*. Here and henceforth,* $\mathcal{P}(\mathbb{S}^{d-1})$ *stands for the set of Borel probability measures on* $\mathbb{S}^{d-1}$*.*

Remark 3.2 (Positional encoding)**.** *For the sake of completeness, in this brief segue we discuss a few ways to perform positional encoding. The original one, proposed in* [VSP+17]*, proceeds as follows. Consider a sequence* $(w_i)_{i\in[n]}\in(\mathbb{R}^d)^n$ *of word embeddings. Then the positional encoding* $p_i\in\mathbb{R}^d$ *of the* i*-th word embedding is defined as* $(p_i)_{2k} = \sin(\frac{i}{M^{2k/d}})$ *and* $(p_i)_{2k+1} = \cos(\frac{i}{M^{2k/d}})$ *for* $k\in[d/2-1]$*, and* $M>0$ *is a user-defined scalar equal to* 10^4 *in* [VSP+17]*. The* i*-th token is then defined as the addition:* $x_i(0) = w_i + p_i$*. Subsequent works simply use either a random*[8] *positional encoding (i.e.,* p_i *is just some random vector) or a trained transformation. The addition can also be replaced with a concatenation* $x_i(0) = [w_i; p_i]$*. (See* [LWLQ22, XZ23] *for details.)*

Remark 3.3 (Mean field limit)**.** *Although the analysis in this paper is focused on the flow of the empirical measure, one can also consider* (3.4) *for arbitrary initial probability measures* $\mu(0)\in\mathcal{P}(\mathbb{S}^{d-1})$*. Both views can be linked through a mean-field limit-type result, which can be shown by making use of the Lipschitz nature of the vector field* $\mathcal{X}[\mu]$*. The argument is classical and dates back at least to the work of Dobrushin* [Dob79]*. Consider an initial empirical measure* $\mu_n(0) = \frac{1}{n}\sum_{i=1}^n \delta_{x_i(0)}$*, and suppose that the points* $x_i(0)$ *are such that* $\lim_{n\to+\infty} W_1(\mu_n(0),\mu(0)) = 0$ *for some probability measure* $\mu(0)\in\mathcal{P}(\mathbb{S}^{d-1})$*. (Here* W_1 *denotes 1-Wasserstein distance—see* [Vil09] *for definitions—.) Consider the solutions* $\mu_n(t)$ *and* $\mu(t)$ *to* (3.4) *with initial data* $\mu_n(0)$ *and* $\mu(0)$ *respectively. Dobrushin's argument is then centered around the estimate*

$$W_1(\mu_n(t),\mu(t)) \leqslant e^{O(1)|t|}W_1(\mu_n(0),\mu(0))$$

for any $t\in\mathbb{R}$*, which in the case of* (3.4) *can be shown without much difficulty (see* [Vil01, Chapter 4, Section 1] *or* [Gol16, Section 1.4.2]*). This elementary mean-field limit result has a couple of caveats. First, the time-dependence is exponential. Second, if one assumes that the points* $x_i(0)$ *are sampled i.i.d. according to* μ_0*, then* $W_1(\mu_n(0),\mu(0))$ *converges to zero at rate* $n^{-\frac{1}{d-1}}$ [Dud69, BLG14]*, which deteriorates quickly when* d *grows. Dimension-free convergence has been established in*

[7]Unless stated otherwise, ∇ and div henceforth stand for the spherical gradient and divergence respectively, and all integrals are taken over $\mathbb{S}^{d-1}$.

[8]This rationale supports the assumption that initial tokens are drawn at random, which we make use of later on.

some cases, for instance by replacing the Wasserstein distance with a more careful choice of metric as in [HHL23, Lac23] *or more generally in* [SFG+12]. *Similarly, the exponential time-dependence might also be improved, as recent works in the context of flows governed by Riesz/Coulomb singular kernels, with diffusion, can attest* [RS23, GLBM24] *(see* [LLF23] *for a result in the smooth kernel case). We do not address this question in further detail here. For more references on this well-established topic, the reader is referred to* [Vil01, Gol16, Ser20] *and the references therein.*

3.2. **The interaction energy.** One can naturally ask whether the evolution in (3.4) admits some quantities which are monotonic when evaluated along the flow. As it turns out, the *interaction energy*

$$\mathsf{E}_\beta[\mu] = \frac{1}{2\beta} \iint e^{\beta\langle x,x'\rangle} \,\mathrm{d}\mu(x)\,\mathrm{d}\mu(x') \tag{3.5}$$

is one such quantity. Indeed,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\mathsf{E}_\beta[\mu(t)] &= \iint \beta^{-1} e^{\beta\langle x,x'\rangle} \,\mathrm{d}\partial_t\mu(t,x)\,\mathrm{d}\mu(t,x') \\ &= \int \mathcal{X}[\mu(t)](x) \cdot \int \nabla\left(\beta^{-1}e^{\beta\langle x,x'\rangle}\right) \mathrm{d}\mu(t,x')\,\mathrm{d}\mu(t,x) \\ &= \int \left\|\mathcal{X}[\mu(t)](x)\right\|^2 Z_{\beta,\mu(t)}(x)\,\mathrm{d}\mu(t,x) \end{aligned} \tag{3.6}$$

for any $t \geqslant 0$ by using integration by parts. Recalling the definition of $Z_{\beta,\mu}(x)$ in (3.3), we see that $e^{-\beta} \leqslant Z_{\beta,\mu}(x) \leqslant e^{\beta}$ for all $x \in \mathbb{S}^{d-1}$. The identity (3.6) therefore indicates that E_β increases along trajectories of (3.4). (Similarly, should $V = -I_d$, the energy E_β would decrease along trajectories.) This begs the question of characterizing the global minima and maxima of E_β, which is the goal of the following result.

Proposition 3.4. *Let $\beta > 0$ and $d \geqslant 2$. The unique global minimizer of E_β over $\mathcal{P}(\mathbb{S}^{d-1})$ is the uniform measure*[9] *σ_d. Any global maximizer of E_β over $\mathcal{P}(\mathbb{S}^{d-1})$ is a Dirac mass δ_{x^*} centered at some point $x^* \in \mathbb{S}^{d-1}$.*

This result lends credence to our nomenclature of the case $V = I_d$ as *attractive*, and $V = -I_d$ as *repulsive*. The reader should be wary however that in this result we are minimizing or maximizing E_β among *all* probability measures on $\mathbb{S}^{d-1}$. Should one focus solely on discrete measures, many global minima appear—these are discussed in Section 9.1—. This is one point where the particle dynamics and the mean-field flow deviate. We now provide a brief proof of Proposition 3.4 (see [Tan17] for a different approach).

Proof of Proposition 3.4. The fact that any global maximizer is a Dirac mass is easy to see. We proceed with proving the rest of the statement. Let $f(t) = e^{\beta t}$. The interaction energy then reads

$$\mathsf{E}_\beta[\mu] = \frac{1}{2} \iint f(\langle x,x'\rangle)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(x').$$

[9] That is, the Lebesgue measure on $\mathbb{S}^{d-1}$, normalized to be a probability measure.

The proof relies on an ultraspherical (or Gegenbauer) polynomial expansion of $f(t)$:

$$f(t) = \sum_{k=0}^{+\infty} \widehat{f}(k;\lambda) \frac{k+\lambda}{\lambda} C_k^\lambda(t)$$

for $t \in [-1,1]$, where $\lambda = \frac{d-2}{2}$, C_k^λ are Gegenbauer polynomials, and

$$\widehat{f}(k;\lambda) = \frac{\Gamma(\lambda+1)}{\Gamma(\lambda+\frac{1}{2})\Gamma(\frac{1}{2})} \frac{1}{C_k^\lambda(1)} \int_{-1}^{1} f(t) C_k^\lambda(t)(1-t^2)^{\lambda-\frac{1}{2}} \, \mathrm{d}t$$

where $C_k^\lambda(1) > 0$ (see [DX13, Section 1.2]). According to [BD19, Proposition 2.2], a necessary and sufficient condition for Proposition 3.4 to hold is to ensure that $\widehat{f}(k;\lambda) > 0$ for all $k \geqslant 1$. To show this, we use the Rodrigues formula [Sze39, 4.1.72]

$$C_k^\lambda(t) = \frac{(-1)^k 2^k}{k!} \frac{\Gamma(k+\lambda)\Gamma(k+2\lambda)}{\Gamma(\lambda)\Gamma(2k+2\lambda)} (1-t^2)^{-(\lambda-\frac{1}{2})} \left(\frac{\mathrm{d}}{\mathrm{d}t}\right)^k (1-t^2)^{k+\lambda-\frac{1}{2}},$$

and the fact that $C_k^\lambda(-t) = (-1)^k C_k^\lambda(t)$ for $t \in [-1,1]$, which in combination with integration by parts yield

$$\int_{-1}^{1} t^\ell C_k^\lambda(t)(1-t^2)^{\lambda-\frac{1}{2}} \, \mathrm{d}t \begin{cases} > 0 & \text{if } \ell \geqslant k \text{ and } \ell - k \text{ is even} \\ = 0 & \text{otherwise .} \end{cases}$$

We conclude by using the power series expansion of f. □

3.3. **A Wasserstein gradient flow proxy.** In view of (3.6), one could hope to see the continuity equation (3.4) as the *Wasserstein gradient flow* of E_β, or possibly some other functional (see the seminal papers [Ott01, JKO98], and [AGS05, Vil09] for a complete treatment). The long-time asymptotics of the PDE can then be analyzed by studying convexity properties of the underlying functional, by analogy with gradient flows in the Euclidean case.

For (3.4) to be the Wasserstein gradient flow of E_β, the vector field $\mathcal{X}[\mu]$ defined in (3.2) ought to be the gradient of the first variation $\delta\mathsf{E}_\beta$ of E_β. However, notice that $\mathcal{X}[\mu]$ is a logarithmic derivative:

$$\mathcal{X}[\mu](x) = \nabla \log \int \beta^{-1} e^{\beta\langle x,y\rangle} \, \mathrm{d}\mu(y). \tag{3.7}$$

(This observation goes beyond $Q = K = I_d$ and $V = \pm I_d$ insofar as $Q^\top K = K^\top Q = \pm V$; see [SABP22, Assumption 1]. Because of the lack of symmetry, it has been shown in [SABP22] that (3.7) is not the gradient of the first variation of a functional.

To overcome this limitation on $\mathbb{R}^d$, thus without layer normalization, [SABP22] propose two ways to "symmetrize" (3.4) that both lead to a Wasserstein gradient flow; see [SABP22, Proposition 2]. We focus here on the simplest one which consists in removing the logarithm in (3.7), or equivalently to removing the denominator in (3.2). This is one point where working on the unit sphere is useful: otherwise, the equation on $\mathbb{R}^d$ without layer normalization (as considered in [SABP22]) is ill-posed for general choices of matrices V, due to the fact that the magnitude of the vector field $\mathcal{X}[\mu]$ grows exponentially with the size of the support of μ. On the contrary, on $\mathbb{S}^{d-1}$ the resulting equation is perfectly well-posed.